

Nate Cirino

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Available May 2027

EDUCATION

Northeastern University

May 2027 | MGPA: 3.6/4.0

B.S. in Computer Science, Systems Concentration

Boston, MA

Coursework: Algorithms & Data Structures, Object-Oriented Design, Computer Systems, Digital Design & Computer Architecture, Computer Networks, Foundations of Software Engineering, Systems Security, Cybersecurity, Foundations of Data Science

EXPERIENCE

Wayfair

Jul 2026 – Present

Software Engineering Co-op

Boston, MA

- Own front-of-house device stack (e-labels, mobile, customer- and associate-facing POS) across 4 retail brands
- Building a supplier-agnostic e-label API unifying 4 suppliers behind one interface, powering new outlet launches
- Shipped POS checkout flows: in-store scan-to-basket customer switching and live associate-entry sync

Wolters Kluwer

Jul 2025 – Jun 2026

Software Engineer

Boston, MA

- Overhauled Relay JMS broker serialization with json-io across 20+ microservices, eliminating timeouts at scale
- Profiled and optimized Expert AI search and filtering paths across 20+ microservices, improving relevance at scale
- Built Java Spring microservice for CME credit issuance inside Expert AI's clinician-facing workflow
- Integrated Expert AI into CME accrual/redemption microservices, expanding GenAI CDS to 250K+ clinicians
- Implemented OIDC/IdP authentication via Azure across 20 enterprise health systems, securing 10K+ clinicians

SculptAI

Jan 2025 – Aug 2026

Software Architect

New York, NY

- Held 99.99% uptime across 1-2M monthly model calls for 2K paying users via AWS WAF, Datadog, and K8s
- Architected Flask backend with RAG pipeline for personalized workout recommendations using Bedrock KB
- Led eng org restructure at \$1M-funded startup, partitioning teams into DevOps, dev, and agentic workflows
- Continue as technical advisor on architecture and agentic-workflow direction

Chevron New Energies

May 2024 – Aug 2024

Software Engineer Intern

Houston, TX

- Built RESTful APIs and data pipelines using Azure Data Factory, improving product efficiency by 40%
- Implemented CI/CD workflows using Azure DevOps and K8s, enabling automated deployment and rollback

PROJECTS

Kepler | Rust, C++, Raft, LSM-tree, MVCC, Concurrency, Tokio

Mar. 2026 – Present

- Built a distributed, linearizable key-value store in Rust with from-scratch Raft consensus and LSM-tree engine
- Implemented Raft election, log replication, and membership changes; storage: memtable, WAL, leveled SSTables
- Added MVCC transaction layer giving snapshot isolation and linearizable reads across a multi-node Raft cluster
- Built single-tier compaction with full SSTable rewrite every 4 flushes; inline, writer-blocking execution

Sentinel | Next.js, FastAPI, Claude, pgvector, OpenTelemetry, Docker

Apr. 2026 – Present

- Built an AI SRE agent autonomously triaging incidents via tool-use over Prometheus, Tempo, and Loki telemetry
- Built pgvector RAG over runbooks and postmortems with prompt caching on frozen schemas; 100% eval accuracy
- Human-in-the-loop remediation gated behind approval; running live in SculptAI production triaging incidents

NOAA AUV Sonar Detection | Python, PyTorch, AWS Neuron SDK, SageMaker

April 2025 – July 2025

- Designed AWS Neuron SDK deployment path for onboard custom-silicon inference under hard real-time limits
- Trained CNN models on thermal sensor spectrograms to detect novel aquatic signatures for NOAA AUVs
- Refined architecture and loss formulation to address class imbalance and sensor noise, improving detection

SKILLS & HONORS

Languages/Tools: C++, C, Rust, Java, Python, TypeScript, JavaScript, Bash, Linux, PostgreSQL, Docker, Kubernetes, AWS, Azure, Terraform, Jenkins, Buildkite, Groovy, GraphQL

Systems: Concurrency & multi-threading, distributed consensus (Raft), MVCC, LSM-tree storage, profiling, real-time systems

Frameworks & Libraries: React, Next.js, Node.js, Express, Flask, PyTorch, TensorFlow, Pandas, FastAPI

AI / Agentic Tooling: Devin, Cursor, Claude Code, Claude Cowork, GitHub Copilot, CodeRabbit

Honors: Dean's List (All Semesters), Tech Lead at Northeastern Electronic Racing, PawHacks, json-io OSS Contributor